

EgoModelKit

Lab Run Guide

After installation

- 1 Run egomodelkit_machine_setup.py (Linux) or Run egomodelkit_machine_setup.bat (Windows)
- 2 Choose Start EgoModelKit.
- 3 Wait for the browser, then follow the pages below.

Run Dry Run before Run Model.

Processing stays on the lab machine.

1. Start analysis

Select Start Analysis.

EgoModelKit



EgoModelKit

Egocentric video analysis for rehabilitation research

Analyze daily-activity videos using computer vision models and generate structured hand-use metrics, timelines, and reproducible research outputs.



Video



Model



Metrics



Research outputs

Local processing. All analysis runs on your computer. No data is uploaded or shared.

Start analysis

2. Choose a workflow

Select Hand interaction or Activity recognition (ADL).

EgoModelKit

- 1 Select model
- 2 Choose input
- 3 Choose output
- 4 Review and run
- 5 Results

Select a model

Choose the workflow you want to run.



Hand interaction

Measures functional hand-object interactions in egocentric videos.

Input: single MP4 video or multiple MP4 videos

Output: interaction profiles and hand-use metrics



Activity recognition (ADL)

Processes egocentric video clips for activity of daily living (ADL) recognition.

Input: single MP4 video or multiple MP4 videos

Output: segment predictions and video/session summaries

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2A. Hand interaction

Select the dominant hand, then Continue.

EgoModelKit

1 Select model

2 Choose input

3 Choose output

4 Review and run

5 Results

Select a model

Choose the workflow you want to run.

☒ **Hand interaction**

Measures functional hand-object interactions in egocentric videos.

Input: single MP4 video or multiple MP4 videos
Output: interaction profiles and hand-use metrics

☐ **Activity recognition (ADL)**

Processes egocentric video clips for activity of daily living (ADL) recognition.

Input: single MP4 video or multiple MP4 videos
Output: segment predictions and video/session summaries

Interaction settings

Dominant hand after injury

Right

Left

Used to label dominant-hand and non-dominant-hand interaction metrics.

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2B. Activity recognition

Select the ADL workflow, then Continue.

EgoModelKit

1 Select model

2 Choose input

3 Choose output

4 Review and run

5 Results

Select a model

Choose the workflow you want to run.

☐ **Hand interaction**

Measures functional hand-object interactions in egocentric videos.

Input: single MP4 video or multiple MP4 videos

Output: interaction profiles and hand-use metrics

☒ **Activity recognition (ADL)**

Processes egocentric video clips for activity of daily living (ADL) recognition.

Input: single MP4 video or multiple MP4 videos

Output: segment predictions and video/session summaries

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[Continue](#)

3. Choose input

Drag in the video(s), or select Choose Input.

EgoModelKit

- ✓ Select model
- 2 Choose input
- 3 Choose output
- 4 Review and run
- 5 Results

Choose input

Select single MP4 video or multiple MP4 videos



Drop input or choose from your computer

Choose input

Supported files: .mp4

No input selected yet.

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Continue

3. Confirm input

Check the selected file(s), then Continue.

EgoModelKit

- ✓ Select model
- 2 Choose input
- 3 Choose output
- 4 Review and run
- 5 Results

Choose input

Select single MP4 video or multiple MP4 videos



Drop input or choose from your computer

Choose input

Supported files: .mp4

Selected: 1 file

- putting_items_in_fridge.mp4

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Continue

4. Choose output folder

Select where EgoModelKit should save the run.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- 3 Choose output
- 4 Review and run
- 5 Results

Choose output folder

Select a folder where EgoModelKit should save the results.



No output folder selected

Choose Output Folder

A new run folder will be created inside the selected output folder.

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4. Confirm output folder

Check the folder, then Continue.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- 3 Choose output
- 4 Review and run
- 5 Results

Choose output folder

Select a folder where EgoModelKit should save the results.



/home/emkclean/Downloads/data/output

Choose Output Folder

A new run folder will be created inside the selected output folder.

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Continue

5A. Review hand-interaction run

Check the summary. Select Dry Run before Run Model.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- 4** Review and run
- 5 Results

Review and run

Confirm the model, input, and output location before starting.

Summary

Model:	Hand interaction
Dominant hand:	Right
Input:	putting_items_in_fridge.mp4
Output folder:	/home/emkclean/Downloads/data/output
Processing mode:	Local

Dry run checks the input, output folder, and local setup without running the full model.

Ready to start.

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Dry Run

Run Model

5A. Hand-interaction dry run

Continue only after all checks pass.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- 4** Review and run
- 5 Results

Review and run

Confirm the model, input, and output location before starting.

Summary

Model:	Hand interaction
Dominant hand:	Right
Input:	putting_items_in_fridge.mp4
Output folder:	/home/emkclean/Downloads/data/output
Processing mode:	Local

Dry run checks the input, output folder, and local setup without running the full model.

✓ Dry run completed successfully.

Checking selected input...

Checking output folder...

Checking local runtime...

✓ Dry run completed successfully.

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Dry Run

Run Model

5A. Run hand-interaction analysis

Keep EgoModelKit open while processing.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- 4 Review and run
- 5 Results

Review and run

Confirm the model, input, and output location before starting.

Summary

Model:	Hand interaction
Dominant hand:	Right
Input:	putting_items_in_fridge.mp4
Output folder:	/home/emkclean/Downloads/data/output
Processing mode:	Local

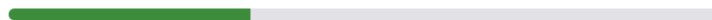
Dry run checks the input, output folder, and local setup without running the full model.

Running model...

Run ID: run-2026-08-05-142354

Extracting frames: 512 / 512 frames
Organizing extracted frames: 512 / 512 frames
Running hand-object contact on extracted frames: 16 / 512 frames
Calculating interaction profiles: 0 / 512 frames
Calculating hand-use metrics: 0 / 512 frames
Saving frame-level predictions: 0 / 512 frames

Overall progress estimate



This may take several minutes. Please keep this window open.

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Cancel Run

6A. Hand-interaction results

Review the summary, metrics, and interaction timeline.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- ✓ Review and run
- 5 Results

Run completed

Your results were saved successfully.

Completed successfully

Model:	Hand interaction
Input:	putting_items_in_fridge.mp4
Dominant hand:	Right
Output folder:	/home/emkclean/Downloads/data/output/run-2026-08-05-142354
Run ID:	run-2026-08-05-142354
Running mode:	Local
Status:	Completed

Clinical hand-use metrics

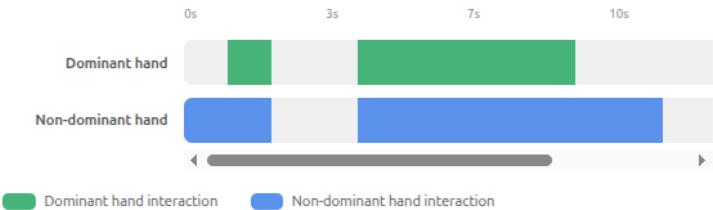
Session-level summary of detected hand-object interactions during the activity.

Metric	Dominant Hand	Non-Dominant Hand	Bilateral / Total
Percent interaction time	35.2%	64.5%	49.8%
Interaction duration	6.0 s	11.0 s	17.0 s
Number of interaction segments	2	3	5

These metrics summarize how often and how long each hand interacted with objects during the analyzed session.

Interaction timeline

Visual overview of when each hand interacted with objects across the analyzed session.



- Open Output Folder
- Start New Run
- View Output Preview

6A. Hand-interaction files

Open the output folder for reports and technical files.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- ✓ Review and run
- 5 Results

Output folder preview

Review what EgoModelKit saved for this run.

Logs and technical files are kept separately for reproducibility and troubleshooting.

Output folder structure

```
> output/  
  > run-2026-08-05-142354/  
    README.txt  
    run_summary.json  
    run_manifest.json  
    > results/  
      video_level_metrics.csv  
      session_level_metrics.csv  
      video_level_metrics_summary.csv  
    > technical/  
      > model_outputs/  
        hand_interaction_input_manifest.csv  
      > post_processing/  
        hand_interaction_subclip_manifest.csv  
        frame_level_predictions.csv  
        interaction_segments.csv  
        metrics_confid.ison
```

What the output folder contains

Most users should review video_level_metrics.csv first. Frame-level outputs and raw model files are stored separately as technical files.

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[Open Output Folder](#)

5B. Review ADL run

Check the summary. Select Dry Run before Run Model.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- 4** Review and run
- 5 Results

Review and run

Confirm the model, input, and output location before starting.

Summary

Model:	Activity recognition (ADL)
Input:	putting_items_in_fridge.mp4
Output folder:	/home/emkclean/Downloads/data/output
Processing mode:	Local

Dry run checks the input, output folder, and local setup without running the full model.

Ready to start.

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Dry Run

Run Model

5B. ADL dry run

Continue only after all checks pass.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- 4** Review and run
- 5 Results

Review and run

Confirm the model, input, and output location before starting.

Summary

Model:	Activity recognition (ADL)
Input:	putting_items_in_fridge.mp4
Output folder:	/home/emkclean/Downloads/data/output
Processing mode:	Local

Dry run checks the input, output folder, and local setup without running the full model.

✓ Dry run completed successfully.

Checking selected input...

Checking output folder...

Checking local runtime...

✓ Dry run completed successfully.

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Dry Run

Run Model

5B. Run ADL analysis

Keep EgoModelKit open while processing.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- 4** Review and run
- 5 Results

Review and run

Confirm the model, input, and output location before starting.

Summary

Model:	Activity recognition (ADL)
Input:	putting_items_in_fridge.mp4
Output folder:	/home/emkclean/Downloads/data/output
Processing mode:	Local

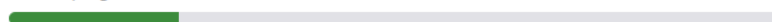
Dry run checks the input, output folder, and local setup without running the full model.

Running model...

Run ID: run-2026-08-05-143331

Extracting frames: 60 / 60 frames
Running object detection model on extracted frames: 5 / 60 frames
Running hand-object contact on extracted frames: 0 / 60 frames
Combining predictions: 0 / 60 frames
Building ADL video and session summaries: 0 / 60 frames

Overall progress estimate



This may take several minutes. Please keep this window open.

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Cancel Run

6B. ADL results

Review the activity timeline and activity summary.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- ✓ Review and run
- 5 Results

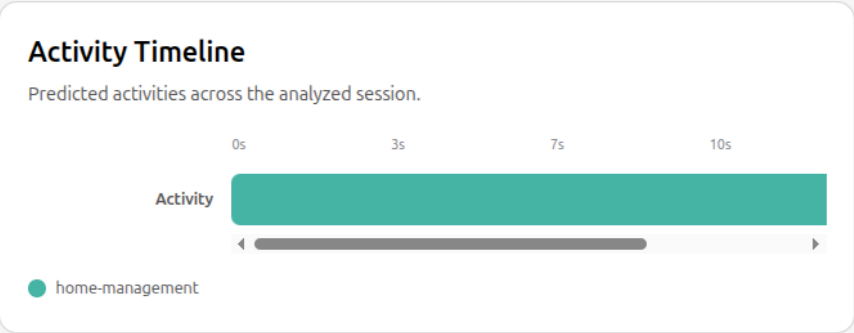
Run completed

Your results were saved successfully.

✓

Completed successfully

Model:	Activity recognition (ADL)
Input:	putting_items_in_fridge.mp4
Output folder:	/home/emkclean/Downloads/data/output/run-2026-08-05-143331
Run ID:	run-2026-08-05-143331
Running mode:	Local
Status:	Completed



Activity Summary

Session-level summary of the activities predicted across all analyzed segments.

ACTIVITY	TOTAL DURATION	SESSION %	SEGMENTS
home-management	17 sec	100.0%	1
Total analyzed session	17 sec	100.0%	1

- Open Output Folder
- Start New Run
- View Output Preview

6B. ADL files

Open the output folder for reports and technical files.

EgoModelKit

- ✓ Select model
- ✓ Choose input
- ✓ Choose output
- ✓ Review and run
- 5 Results

Output folder preview

Review what EgoModelKit saved for this run.

Logs and technical files are kept separately for reproducibility and troubleshooting.

Output folder structure

```
> output/  
  > run-2026-08-05-143331/  
    README.txt  
    run_summary.json  
    run_manifest.json  
    > results/  
      adl_segment_predictions.csv  
      adl_video_summary.csv  
      adl_session_summary.csv  
    > technical/  
      > model_outputs/  
        adl_input_manifest.csv  
        all_preds.pkl  
      > post_processing/  
        adl_segment_manifest.csv  
        adl_processing_config.json  
    > intermediate_files/
```

What the output folder contains

Review adl_segment_predictions.csv first. The video and session files are descriptive ADL summaries; they do not contain hand-use metrics. Files under technical/ include binary model artifacts that are not intended to be opened as text.

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Open Output Folder